

Imbalanced Sample Selection With Deep Reinforcement Learning for Fault Diagnosis

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Abstract—An imbalanced number of faulty and normal samples causes serious damage to the performance of the conventional diagnosis methods. To settle the data-imbalance fault diagnosis problem, this article presents a novel general imbalanced sample selection strategy (DiagSelect) based on deep reinforcement learning. In DiagSelect, the problem of imbalanced sample selection from the training set is formulated as a multiarmed bandit problem of deep reinforcement learning. The nondifferentiable optimization problem of imbalanced sample selection can be solved by the Markov decision process. The parameters of DiagSelect can be optimized by REINFORCE with the feedback of the validation set. DiagSelect performs intelligent imbalanced sample selection to obtain better diagnosis performance autonomously. As a data-level technique, DiagSelect can be easily used in conjunction with the conventional diagnosis models. DiagSelect is validated in a synthetic dataset and an industrial process dataset. The results have shown the effectiveness, stability, and transferability of DiagSelect.

Index Terms—Deep reinforcement learning, fault diagnosis, imbalanced sample selection, industrial process monitoring, multi-armed bandit (MAB) problem, nondifferentiable optimization problem.

I. INTRODUCTION

WITH the rapid development of the Internet of Things and cloud computing, a large amount of industrial process data has been recorded in the industrial distributed control system [1]. In the context of smart manufacturing of

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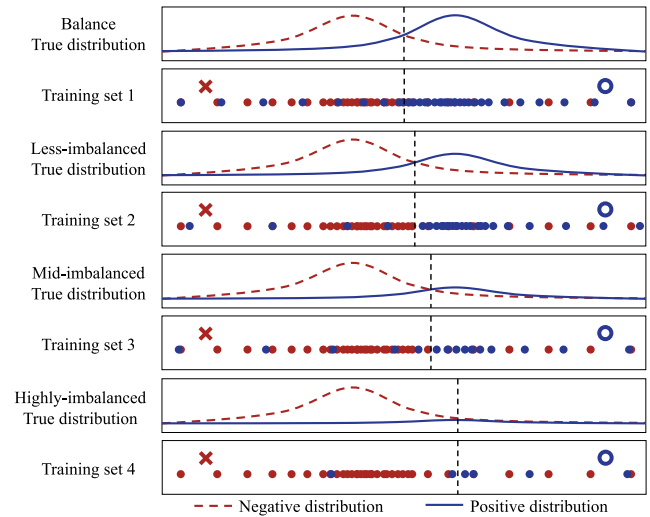


Fig. 1. Influence illustration of different imbalance levels.

industry 4.0, various new intelligent technologies are used for process monitoring [2], control [3], and quality prediction [4]. Data-driven fault diagnosis is an important part of industrial process monitoring, which distinguishes faults from normal process data by using easy-to-measure process variables. Data-driven fault diagnosis methods have been extensively studied because of their flexibility, ease of use, and low cost. In recent years, many fault diagnosis methods based on machine learning have been proposed and applied to various industrial processes [5]–[7].

As the number of industrial process samples collected continues to increase in modern factories [8], the industrial database contains more and more information. Most industrial processes operate under normal conditions, the collected faulty samples are rare. As a result, the number of normal samples collected in the industrial process is far greater than that of faulty samples. Because there are only relatively few faulty samples, it is difficult for conventional fault diagnosis methods to learn effective information. Such problems are defined as the class-imbalance problem in the fault diagnosis domain. As shown in Fig. 1, industrial process data is divided into four levels of imbalance: balance distribution, less-imbalanced distribution, mid-imbalanced distribution, and highly imbalanced distribution. The four levels of imbalance lead to different degrees of inaccurate model fitting. Thus, there is an urgent need to improve the performance of

fault diagnosis methods without excessively increasing expert knowledge and domain knowledge. To settle such an issue, a lot of research works have been carried out, which can be divided into three categories as follows:

- 1) *Data-Level Techniques*: Data-level techniques tend to rebalance class distribution through various data sampling or generating techniques in the data preprocessing phase. The data-level techniques are mainly divided into four categories: under-sampling [9], over-sampling [10], [11], cleaning-resampling [12], and data generation [13]. However, under-sampling and over-sampling often lead to over-fitting and under-fitting without any guidance. Cleaning-resampling needs to meet some assumptions. As a novel data-level technique, data generation has been studied extensively, and it is effective but unstable.
- 2) *Algorithm-Level Methods*: The algorithm-level methods mainly solve the class-imbalance problem directly from the perspective of algorithm design. Cost-sensitive learning is an important part of algorithm-level methods, which includes adapting learning algorithm [14] and weighted data space [15]. Cost-sensitive learning is versatile in many domains [16]. However, it requires domain experts to provide the cost matrix in advance, which is usually not feasible in practice. Many recent algorithm-level methods are mainly based on designing new loss functions, such as FocalLoss [17].
- 3) *Hybrid Approaches*: Hybrid approaches are the combination of data-level techniques and algorithm-level methods, which effectively improve effective solutions for the class-imbalance problem in the fault diagnosis domain [18].

In the field of fault diagnosis, these class-imbalance methods have good effects on the class-imbalance problem. However, the above methods have three main limitations: 1) the performance is unstable because they are sensitive to outliers; 2) the applicability is poor because the hand-designed cost matrix requires expert knowledge as the prerequisite; and 3) the adaptive ability is poor because these methods cannot be applied to all imbalanced industrial process datasets. Therefore, these methods are not general and adaptive for complex industrial process tasks without any expert knowledge and assumption.

To settle the above problems, it is necessary to propose an adaptive automatic sample selection strategy that can automatically remove outliers and select effective samples to reduce the degree of imbalance. To realize this objective, a new adaptive automatic imbalanced sample selection strategy called DiagSelect is proposed in this work. The design of DiagSelect is inspired by the latest developments of reinforcement learning, which formalizes the problem of removing outliers and selecting effective samples as a MAB problem. In DiagSelect, the reinforcement learning mechanism is used to solve this MAB problem. The REINFORCE [19] is used to optimize the process of sample selection. More specifically, the key idea of DiagSelect is that process of sample selection is trainable and real-time interaction

with the collected samples by several time steps. As a MAB problem, the process of sample selection in the training set is described as a single-state Markov decision process (MDP). At each time step, the state is the training set, which includes feature information and label information. The reward is the performance of the diagnosis model of DiagSelect on the validation set after training on the selected sample subset. The diagnosis model of DiagSelect is just to analog the environment in reinforcement learning for obtaining the reward. The action is selecting samples from the training set. The goal of the entire selection process is to maximize the benefits of the sample selection by several time steps. To summarize, the main contributions of this article are summarized as follows:

- 1) From a new perspective, the novel strategy automatically learns the adaptive imbalanced sample selection model for different domains datasets. DiagSelect can settle the class-imbalance problem of a variety of distribution adaptively without any expert knowledge and assumption in the fault diagnosis domain.
- 2) The adaptive automatic process of the imbalanced sample selection is first formalized as the MDP. The problem of finding the best sample subset is formalized as the MAB problem.
- 3) The proposed strategy is evaluated in a synthetic dataset and an industrial process dataset. These experiments and comparative studies validate the effectiveness of tailored designs of DiagSelect for the class-imbalance problem in the fault diagnosis domain. DiagSelect is decoupled from specific diagnosis models, which makes DiagSelect universally applicable to most statistical and nonstatistical diagnosis models [such as decision tree (DT), support vector machine (SVM), and multilayer perceptron (MLP)].

In the case study, random under-sampling (RUS), random over-sampling (ROS), synthetic minority oversampling technique (SMOTE), ADaptive SYNthetic over-sampling (ADASYN), SVMSMOTE, SMOTE with Tomek links cleaning (SMOTETomek), and EasyEnsemble are selected as comparison strategies. Meanwhile, three diagnosis models that include DT, SVM, and MLP are selected to verify the generalization of the DiagSelect strategy. The results of a synthetic dataset and an industrial process dataset show that the DiagSelect-based diagnosis model can be utilized in the industrial process data for the class-imbalance problem, which has a strong diagnosis ability and stability under different IRs. The cross-task experiment shows that DiagSelect has transferability in different datasets and different diagnosis models. This transferability allows DiagSelect to be used directly or initially for another task to reduce the calculation costs. Consequently, DiagSelect is an effective strategy for the class imbalance problems in the field of fault diagnosis.

The rest of this article is organized as follows. In Section II, a review of relative preliminary work is briefly introduced. In Section III, the overall framework and details of the proposed DiagSelect are presented. In Section IV, the effectiveness, stability, and transferability of DiagSelect are verified by comparing it

with other strategies in two cases. Finally, Section V concludes this article.

II. RELATED WORK

A. Fault Diagnosis With Imbalanced Data

Since the collected data usually contains a small number of faulty samples and many normal samples, the implementation of fault diagnosis with imbalanced data is important for machine health management and industrial process monitoring [20]. The existing research on fault diagnosis with imbalanced data is mainly divided into three categories: data augmentation-based strategy, feature learning-based strategy, and model design-based strategy [21]. Data augmentation-based strategy obtains better diagnosis performance by generating synthetic samples, oversampling existing samples, and reweighting samples. Yi *et al.* [22] proposed a minority clustering SMOTE (MC-SMOTE) method, which involves clustering minority samples to improve imbalanced classification performance. After the data augmentation-based strategy, the processed data can be directly used in the conventional fault diagnosis models. Feature learning-based strategy can ameliorate the diagnosis performance by designing regularized neural networks or feature adaptation to extract features from limited fault data. Model design-based strategy gains high diagnosis performance by directly designing appropriate fault diagnosis model structure [23]. In addition, when the minority samples are extremely rare, some knowledge-driven methods are proposed with limited sample data. For example, Li *et al.* [24] proposed a novel meta learning fault diagnosis method, which acquires prior knowledge to achieve fast and accurate few-shot bearing fault diagnosis by meta learning. Zhuo *et al.* [25] proposed a generative model with fault attribute space, which is optimized by the auxiliary triplet loss to the any-shot problem with limited sample data. In this work, deep reinforcement learning is considered to solve the class-imbalance problem in the fault diagnosis domain for the first time.

III. DIAGSELECT

In this section, the first part introduces the overall flowchart of using DiagSelect for fault diagnosis, which includes how to train DiagSelect and identify fault types online. The second part introduces the framework of DiagSelect, which includes the detailed formulation, the training procedure, and definitions of state, action, and reward. The third part introduces the detailed optimization for the formulation of DiagSelect. The last part introduces the detailed procedure of online testing of DiagSelect.

A. Overall Flowchart

The overall flowchart of DiagSelect for fault diagnosis mainly includes offline training and online testing, which contains five steps as follows:

- Offline Training

Step 1: Preprocess Dataset. The z-score standardization [26] is performed in the offline dataset and online dataset. Meanwhile,

the offline dataset is split into the training set and validation set. The online dataset is used as the test set.

Step 2: Pretrain the agent of DiagSelect. In the pretraining phase, the agent of DiagSelect is trained by the cross-entropy loss function based on the original training set with initial probability. The output of the agent is the probability of whether the sample is selected.

Step 3: Train the agent of DiagSelect. The sample selection problem in DiagSelect is formalized as a MAB problem, which is equivalent to the single-state problem in reinforcement learning. In the offline training phase, the REINFORCE loss of policy gradient in reinforcement learning is used for training the agent. The state is all samples in the training set, which includes feature information and label information. The action is to select samples from the training set. The reward is the performance of the diagnosis model of DiagSelect on the validation set, which is trained on the selected sample subset. At each time step, first, the sample subset is selected from the training set by the agent of DiagSelect. Second, the diagnosis model of DiagSelect is trained on the selected sample subset. Third, the reward is obtained by evaluating the diagnosis performance of the diagnosis model of DiagSelect on the validation set. Finally, the parameters of the agent are updated by REINFORCE loss with the reward and the probability that the sample is selected. The time step runs until the agent of DiagSelect converges. More detailed training strategies will be mentioned in Section III-B.

- Online Testing

Step 4: Obtain online fault diagnosis model. DiagSelect is a strategy for sample selection, which is decoupled from the online fault diagnosis model. After the offline training phase, the agent of DiagSelect is obtained. The sample subset can be selected from the training set by the agent of DiagSelect. The online fault diagnosis model trained on the selected sample subset can be employed to recognize the fault type of online samples.

Step 5: Online fault diagnosis. When a new sample arrives, the online fault diagnosis model can be applied to the online diagnosis task to recognize the fault type of the new sample.

The flowchart of DiagSelect applied to the fault diagnosis task is depicted in Fig. 2.

B. Model Framework

In this section, some details of model framework are introduced. Given the imbalanced dataset $\mathcal{D} = \{\mathbf{x}_k, \mathbf{y}_k\}_{k=1}^N$, where $N = \sum_{i=0}^{C-1} n_i$ is the number of samples, n_i is the number of samples from class i , and C is the number of classes. In general, the subscripts of the class are sorted in descending order of the number of samples in each class, e.g. if $i < j$, then $n_i > n_j$, and $n_0 \gg n_{C-1}$. Let $\mathcal{D}_{\text{tm}} = \{X_{\text{tm}}, Y_{\text{tm}}\}$ be the training set, $\mathcal{D}_{\text{val}} = \{X_{\text{v}}, Y_{\text{v}}\}$ be the validation set, and $\mathcal{D}_{\text{tst}} = \{X_{\text{tst}}, Y_{\text{tst}}\}$ be the test set. Imbalance ratio (IR) = $|\mathcal{N}|/|\mathcal{P}|$, where $|\mathcal{N}|$ is the sample size of the majority class and $|\mathcal{P}|$ is the sample size of the minority class. The detail training framework of DiagSelect is presented in Fig. 3.

To enable the model converge more stable, the training set $\mathcal{D}_{\text{tm}}$ is used to pretrain the agent of DiagSelect. In the pretraining

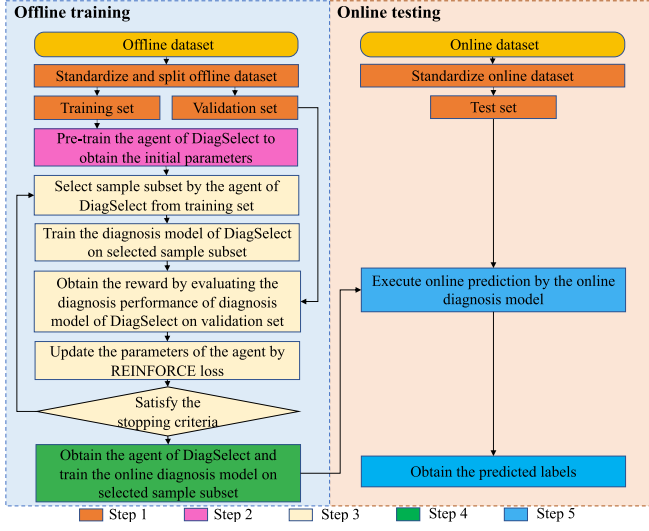


Fig. 2. Flowchart of DiagSelect applied to the fault diagnosis in the industrial process.

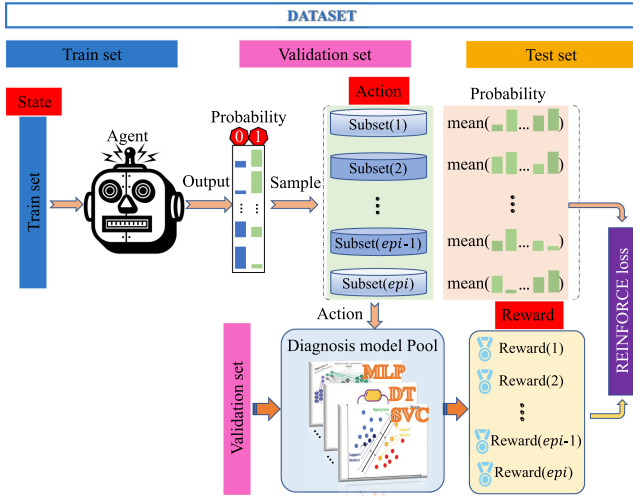


Fig. 3. Model training framework of DiagSelect.

phase, the input of agent of DiagSelect is $X_{\text{trn}} \oplus \text{onehot}(Y_{\text{trn}})$ and the loss of agent is defined as follows:

$$\mathcal{L}_{\text{pre}} = -\frac{1}{|\mathcal{D}_{\text{trn}}|} \sum_{i=0}^{|\mathcal{D}_{\text{trn}}|-1} p_{\text{ini}}^i \log p_i \quad (1)$$

where $|\mathcal{D}_{\text{trn}}|$ represents the sample size of $\mathcal{D}_{\text{trn}}$, p_i is the 2-D output of agent, and p_{ini}^i is the 2-D initial probability of sample i . The first dimension of p_{ini}^i is the probability that sample i is not selected, and the second dimension of p_{ini}^i is the probability that sample i is selected. To enable the agent network to converge better, the second dimension of p_{ini}^i is the proportion of the sample size of the corresponding class to the number of all samples. The sum of the values from the first dimension and the second dimension of p_{ini}^i is one. After the pretraining phase, the naive DiagSelect has been obtained. For the strong DiagSelect, reinforcement learning is introduced to further optimize DiagSelect.

We need to define π_θ , a_t , and r_t as the agent, action, and reward at time t , respectively. In the framework of DiagSelect, we formulate the sample selection as the action a_t from the agent π_θ . In the MAB problem, each action is selected with an expectation or average reward. The expectation reward of each action a_t is defined as $q_*(a_t)$

$$q_*(a_t) \doteq \mathbb{E}[r_t | a_t]. \quad (2)$$

This work adopts the average sampling method, which evaluates the reward of each action by calculating the average reward. The average reward is described as follows:

$$Q(a_t) \doteq \frac{1}{n} \sum_{i=1}^n r_i. \quad (3)$$

The goal of the MAB problem is to find the action corresponding to the maximum $Q(a_t)$

$$A_t \doteq \arg \max_a Q(a_t). \quad (4)$$

Based on the MAB problem, we make a key conceptual assumption that formulates the sequential process of iterating the sample selection as an MDP. Formally, the MDP can be defined as the tuple $M = \{S, A, R\}$, where S is the state, A is the action, and R is the reward. More specifically, all components are defined as follows.

1) **State:** S is the state from the tuple M of MDP. $S = \{s_t | t = 0, \dots, T\}$, T is the max value of the time step. The state s_t is represented by the concatenation of process variables and class labels from the training set at time step t . The state s_t is defined as follows:

$$s_t = X_{\text{trn}} \oplus \text{onehot}(Y_{\text{trn}}) \quad (5)$$

where X_{trn} is the process variables of training set and Y_{trn} is the class labels of the training set. At each time step, the sample order of training set is shuffled.

To encode s_t for reinforcement learning, the gated recurrent unit (GRU) is selected to generate a dense vector representation h_t . The mechanism of GRU makes the state of the agent have relevance in multiple time steps of the agent, which makes it unnecessary to manually define the changing and appropriate states. The details of the encoding process are defined as follows. The update ratio z_t is

$$z_t = \sigma(W_z s_t + U_z h_{t-1}) \quad (6)$$

where s_t is the state of current time step, h_{t-1} is the dense vector representation of state s_{t-1} at the previous time step, W_z is the input weights of update gate, U_z is the output weights of update gate, and $\sigma(\cdot)$ denotes the sigmoid function. The reset ratio o_t is

$$o_t = \sigma(W_o s_t + U_o h_{t-1}) \quad (7)$$

where W_o is the input weights of update gate, and U_o is the output weights of update gate. The intermediate variable $\tilde{h}_t$ is

$$\tilde{h}_t = \tanh(W s_t + U(o_t \odot h_{t-1})) \quad (8)$$

where W is the input weights, U is the output weights, and $\odot$ is the Hadamard product. The dense vector representation h_t of

state s_t at the current time step is

$$h_t = z_t \odot h_{t-1} + (1 - z_t) \odot \tilde{h}_t \quad (9)$$

where z_t is the update ratio, h_{t-1} is the dense vector representation of state s_{t-1} at the previous time step, and $\tilde{h}_t$ is the intermediate variable.

2) **Action:** A is the action from the tuple M of MDP, which is obtained by the agent π_θ . In the framework of the DiagSelect, the action is to select samples from the training set $\mathcal{D}_{\text{trn}}$. The output of the agent π_θ is 2-D. The first dimension is the probability that the sample is not selected, and the second dimension is the probability that the sample is selected.

After obtaining the dense vector representation h_t of current state s_t , the probability that the sample is selected should be calculated. To obtain the action probability at time step t , we feed the dense vector representation h_t into the fully connected layer

$$d_t = h_t W_f^T + b_t \quad (10)$$

$$\pi_\theta(a_{t,i}|s_t) = \frac{\exp(-d_{t,i})}{\sum_{j=1}^C \exp(-d_{t,j})} \quad (11)$$

where W_f and b_t are weights and bias of the fully connected layer. C is the class number, θ is the trainable parameters of the agent π_θ . $d_{t,i}$ is the i th element of d_t . $\pi_\theta(a_{t,i}|s_t)$ is the i th element of $\pi_\theta(a_t|s_t)$. $a_{t,i}$ is the i th element of a_t . When $a_{t,i}$ is one, the sample i is selected. The action is sampled from the agent of $\pi_\theta(a_t|s_t)$ as follows:

$$a_t, p_t = \text{sample}(\pi_\theta(a_t|s_t)) \quad (12)$$

where a_t indicates the 1-D binary vector, zero represents the sample is not selected and one represents the sample is selected. p_t represents the probability of selecting a_t .

3) **Reward:** For the MAB problem, the goal of the formulation is to obtain the best actions and maximize the corresponding rewards. For the agent of DiagSelect, the goal of the formulation is to select the best sample subset. A direct and effective strategy is to train a diagnosis model of DiagSelect on selected samples subset and use the diagnosis performance in the validation set $\mathcal{D}_{\text{val}}$ as the reward. The diagnosis model of DiagSelect is just to analog the environment in reinforcement learning for obtaining the reward. The evaluation criterion is used to be the quantitative indicator for the performance of the diagnosis model of DiagSelect in the validation set $\mathcal{D}_{\text{val}}$. The evaluation criterion of the diagnosis model in DiagSelect is preferably consistent with that of the online fault diagnosis model for online testing. We train the diagnosis model of DiagSelect based on the selected samples subset and obtain the predicted labels $\hat{Y}_v^t$ of the validation set X_v at time step t as follows:

$$\hat{Y}_v^t = f_{\text{dig}}(X_v; D_{\text{trn}}^\dagger) \quad (13)$$

where $D_{\text{trn}}^\dagger$ is the selected samples subset from training set $\mathcal{D}_{\text{trn}}$ when the element of a_t is one, f_{dig} is the diagnosis model of DiagSelect. Then, the reward function in the environment is defined as follows:

$$r_t = \Re(Y_v, \hat{Y}_v^t) \quad (14)$$

where $\Re$ is the evaluation criterion and r_t is the reward at time step t .

C. Optimization of the Markov Decision Process

After defining all components of the MDP of RL, it is necessary for DiagSelect to select the loss function to solve the entire optimization problem. In this work, the REINFORCE loss [19] of the standard policy gradient method is used to solve the entire optimization problem. The loss function is defined as follows:

$$\mathcal{L}(\pi_\theta) = -\sum_{k=1}^{\text{epi}} r_t^k p_t^k. \quad (15)$$

In this loss function, when the agent of DiagSelect makes a backpropagation at time step t , action a_t needs to be collected by epi times to obtain r_t^k and p_t^k . To learn the optimal policy $\theta^{\text{opt}} = \arg \max_{\theta} \mathcal{L}(\pi_\theta)$, the optimization process of $\mathcal{L}(\pi_\theta)$ can be derived as

$$\theta \leftarrow \theta - \rho \sum_{k=1}^{\text{epi}} r_t^k p_t^k \quad (16)$$

where ρ is the learning rate, r_t^k is the reward for collecting the k time at time step t , p_t^k is the probability of a_t^k at time step t .

D. Online Testing

After the agent of DiagSelect has converged, the deterministic action can be obtained from the agent. When the training set $\mathcal{D}_{\text{trn}}$ is fed into the agent, the action can be obtained as follows:

$$a^* = \arg \max_a \pi_\theta(a|s) \quad (17)$$

where $s = X_{\text{trn}} \oplus \text{onehot}(Y_{\text{trn}})$. The online diagnosis model $f_{\text{dig}}^{\text{online}}$ is trained based on the selected samples subset and then the predicted labels $\hat{Y}_{\text{tst}}$ of test set $\mathcal{D}_{\text{tst}}$ can be obtained as follows:

$$\hat{Y}_{\text{tst}} = f_{\text{dig}}^{\text{online}}(X_{\text{tst}}; D_{\text{trn}}^\dagger) \quad (18)$$

where $D_{\text{trn}}^\dagger$ is the selected samples subset of training set $\mathcal{D}_{\text{trn}}$ when the element of a^* is one.

The entire process of DiagSelect includes two parts: offline training and online testing. The pseudocode of offline training of DiagSelect is shown in Algorithm 1. The pseudocode of online testing of DiagSelect is shown in Algorithm 2.

IV. CASE STUDIES

A. Case Study in Synthetic Datasets

1) **Setup Details:** In this case study, we create a series of toy imbalanced datasets from the TwoMoon dataset [27] with different IRs. In our experiment, the toy dataset contains two classes. The number of samples in the minority class is 50, and the number of samples in the majority class varies with the IRs. To evaluate the effectiveness of the proposed DiagSelect for fault diagnosis, it is compared with several strategies under different IRs, as shown in Fig. 4. Comparison strategies come from four major representative branches (under/over-sampling, cleaning-resampling, and ensemble). After the process of under/over-sampling, the number of samples in each class is the same. The selected comparison strategies are as follows: RUS, ROS, SMOTE [10], ADASYN [11], SVMSMOTE [28], SMOTE with

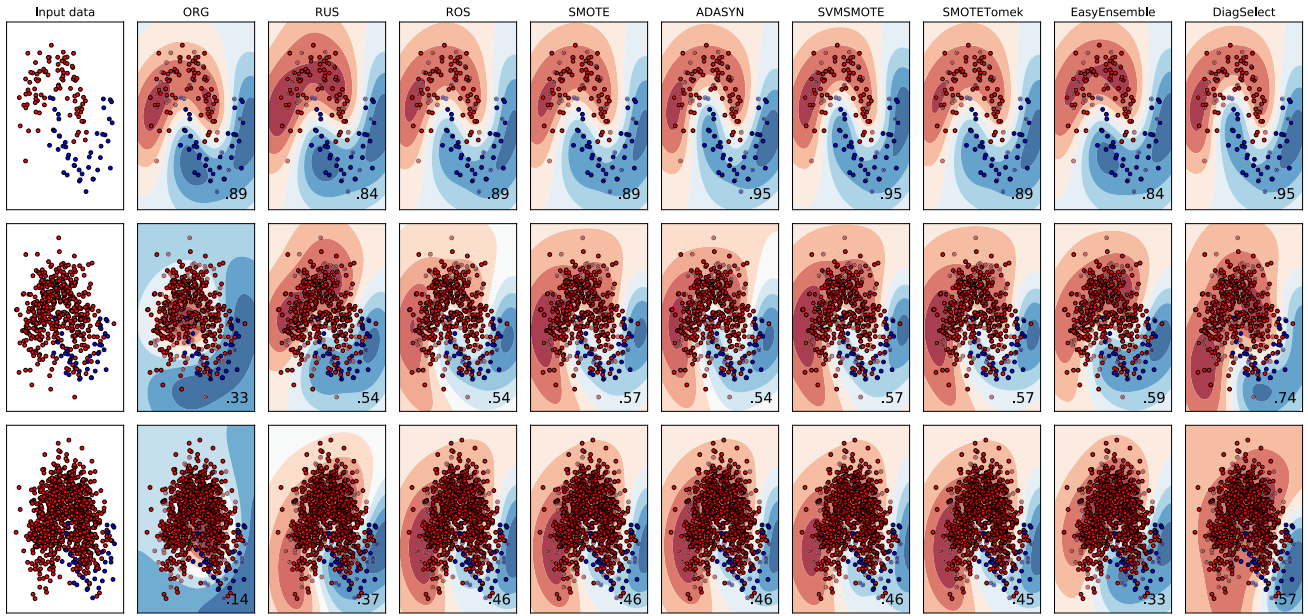


Fig. 4. Comparisons of DiagSelect with other different strategies. The IRs in the 1st, 2nd, and 3rd rows are 2, 10, and 20, respectively. The number in the lower right of each subfigure is F_1 -score.

Tomek links cleaning (SMOTETomek) [12], and EasyEnsemble [29]. ORG represents the original data without any strategy. In this manner, F_1 -score is used as the evaluation criterion. We introduce SVM [30] as a diagnosis model of all strategies. The input dimension of the agent of DiagSelect is 2. The layer number of GRU is 1. The hidden dimension of GRU is 5. The output dimension of the agent of DiagSelect is 2. The optimizer is RMSprop. The learning rate of the RMSprop optimizer is 0.001. The *epi* is 10. The early stop is set to 100.

2) Visualization and Analysis: The class decision boundaries determined by the different strategies based on SVM are plotted in Fig. 4. In general, it can be observed that the diagnosis performance of all strategies-based methods deteriorates with the IR increasing, especially the ORG-based SVM method. In the dataset with IRs (2, 10, and 20), the F_1 -scores of the DiagSelect-based SVM methods are 0.95, 0.74, and 0.57, respectively, which shows that DiagSelect-based SVM methods outperform other strategies-based SVM methods under different IRs. More specifically, when the IR is 2, all strategies-based SVM methods can learn a clear class decision boundary. Among them, the ADASYN-based, SVMSMOTE-based, and DiagSelect-based SVM methods obtain the best performance with the F_1 -score of 0.95. As IR increases from 2 to 10, the performance of all strategies-based SVM methods has deteriorated. The ORG-based SVM method deteriorates the most, which indicates that the imbalance problem destroys the data distribution and makes the decision boundary of the diagnosis model biased. Other strategies-based SVM methods have alleviated the impact of the imbalanced problem to a certain extent. However, the DiagSelect-based SVM method obtains the best performance with the F_1 -score of 0.74, which reflects the effectiveness of the DiagSelect-based SVM method by selecting effective samples and removing redundant samples for imbalanced problems.

When the IR is 20, since the information of the majority class is far more than the information of the minority class, all methods obtain the worst diagnosis performance. Note that even under the highly imbalanced ratio, the DiagSelect-based SVM method can still fit the data distribution and obtain a reasonable and stable decision boundary. This is because the DiagSelect-based SVM method can adaptively select effective samples based on the validation rewards to obtain good diagnosis performance. It can be found that the DiagSelect-based SVM method has a strong generalization ability under different IRs. All results indicate that our DiagSelect-based SVM method is superior to other strategies-based SVM methods in handling imbalanced diagnosis problems.

B. Case Study in the USM Diagnostics

To verify the effectiveness and stability of the proposed DiagSelect in practical applications, we conducted experiments on real-world fault diagnosis tasks. The USM diagnostics datasets are acquired from the UCI machine learning repository [31] and provided by the National Engineering Laboratories (NEL), U.K.. In this work, Meter C and Meter D datasets are selected from the USM diagnostics datasets for this case study. Meter C and Meter D datasets contain four classes, which are healthy, gas injection, installation effects, and waxing. These datasets have 43 features, including profile factor, symmetry, crossflow, flow velocity (in each of the four paths), speed of sound (in each of the four paths), signal strength (at both ends of each of the four paths), signal quality (at both ends of each of the four paths), gain (at both ends of each of the four paths), and transit time (at both ends of each of the four paths). The information of the USM diagnostics datasets is summarized in Table I.

Algorithm 1: Offline training of DiagSelect.

Input: $\mathcal{D}_{trn}$ is the training set. $\mathcal{D}_{val}$ is the validation set. π_θ is the agent. T is the maximum number of time steps. epi is a hyperparameter.

- 1 Pre-train the agent following Eq.1.
- 2 Initialize s_0 based on $\mathcal{D}_{trn}$ following Eq.5.
- 3 Initialize the dense vector representation h_0 to zero.
- 4 **for** $t = 1, \dots, T$ **do**
- 5 Feed the current state s_t and the dense vector representation h_{t-1} into the agent π_θ following Eq.6.
- 6 Obtain the the current dense vector representation h_t following Eq.9.
- 7 Calculate the probability $\pi_\theta(a_t|s_t)$ by the agent π_θ following Eq.11.
- 8 **for** $k = 1, \dots, epi$ **do**
- 9 Sample the a_t^k and p_t^k from $\pi_\theta(a_t|s_t)$ following Eq.12.
- 10 Select samples subset $D_{trn}^\dagger$ by a_t^k .
- 11 Obtain the predicted labels $\hat{Y}_v^t$ following Eq.13.
- 12 Calculate the reward r_t^k following Eq.14.
- 13 **end**
- 14 Calculate the REINFORCE loss of agent π_θ following Eq.15.
- 15 Update the parameters θ of agent π_θ of Eq.16.
- 16 Shuffle sample order of $\mathcal{D}_{trn}$ randomly to obtain a new state s_{t+1} following Eq.5.
- 17 **end**

Output: π_θ

TABLE I
INFORMATION OF USM DIAGNOSTICS DATASET

Dataset	#Samples	#Features	#Classes	IR	Class distribution
Meter C	181	43	4	2.348	54/23/54/50
Meter D	180	43	4	2.391	51/23/55/51

Algorithm 2: Online Testing of DiagSelect.

Input: $\mathcal{D}_{trn}$ is the training set. $\mathcal{D}_{tst}$ is the test set.

- 1: Obtain state s based on $\mathcal{D}_{trn}$ following Eq.5.
- 2: Obtain the agent π_θ by running Algorithm 1.
- 3: Feed the state s into the agent π_θ .
- 4: Obtain the action a^* based on state s following Eq.17.
- 5: Select samples subset $D_{trn}^\dagger$ by the action a^* .
- 6: Train the f_{dig}^{online} on the selected samples subset $D_{trn}^\dagger$ and obtain the predicted labels $\hat{Y}_{tst}$ following Eq.18.

Output: $f_{dig}^{online}, \hat{Y}_{tst}$

1) *Setup Details:* The experimental results are acquired by the mean of four-fold stratified cross-validation. The input dimension of the agent of DiagSelect is 47. The layer number of GRU is 1. The hidden dimension of GRU is 80. The output dimension of the agent of DiagSelect is 4. The optimizer is RMSprop. The learning rate of the RMSprop optimizer is 0.001.

The epi is 10. The early stop is set to 100. The setting of the comparison methods is the same as that of the synthetic dataset. We introduce three canonical diagnosis model (DT [32], SVM [30], and MLP [33]) to validate the effectiveness and stability of DiagSelect-based methods. The evaluation criteria are Balanced Accuracy (bAcc), Macro- F_1 , and G-Mean (GM).

2) *Result Analysis:* To make the results more convincing, the experiment was repeated ten times. Tables II and III show the mean and standard deviation of the diagnosis results of all strategies (i.e., RUS, ROS, SMOTE, ADASYN, SVM SMOTE, SMOTETomek, EasyEnsemble, and DiagSelect) combined with three different canonical diagnosis models. Meanwhile, to reflect the stability of all methods, the boxplot of Macro- F_1 is shown in Figs. 5 and 6. In the fault diagnosis domain, the performance of the strategy varies with the different datasets and diagnosis models. Few methods have the adaptive ability in related fault diagnosis tasks. It can be seen from Tables II and III that the RUS-based method performs worst in the Meter C and Meter D datasets with all diagnosis models. In the case of the RUS-based method, if a relatively large number of samples are deleted from the majority class, it is possible to delete valuable information and change the distribution of the majority class or the entire dataset. In addition to the RUS-based method, other methods have played a role in dealing with the problem of class-imbalance problem in the fault diagnosis domain when combining with specific diagnosis models. However, these methods cannot perform well combined with different diagnosis models in the two datasets. For example, EasyEnsemble works well combined with DT, SVM, and MLP in the Meter D dataset but fails to boost the performance combined with SVM in the Meter C dataset. The reason is that the EasyEnsemble-based method can make the diagnosis model stable, but not the performance of all methods in all datasets can be improved by the EasyEnsemble strategy. Meanwhile, SMOTETomek can perform well combined with DT and SVM but does not perform well combined with MLP. The reason is that they perform model-agnostic resampling/selecting without considering the differences in the diagnosis models. Therefore, it is important to make the model more general by the effective sample selection strategy, which considers the specific diagnosis model and different data distribution adaptively for the class-imbalance problem in the fault diagnosis domain. In the comparative experiment, it can be seen that DiagSelect performs best when used in combination with different models from the perspective of mean and standard deviation in both datasets. Compared to other methods in the Meter C and Meter D datasets, the mean and standard deviation of all evaluation criteria for DiagSelect is higher and smaller than other methods. DiagSelect performs relatively well and is stable in ten independent runs.

3) *Cross-Task Transferability of the DiagSelect:* It can be seen from Table I that the Meter C and Meter D datasets have the same data structure, including the same number of features and classes. Meanwhile, the sample size of each class is also similar. The transferability of DiagSelect is considered to be verified in the Meter C and Meter D datasets. In the transferability experiment, we conducted the experiment on the transferability of cross-model and cross-dataset by the train-test

TABLE II
DIAGNOSIS PERFORMANCE BY DIFFERENT METHODS IN METER C DIAGNOSTICS DATASET

Dataset	Model	Metric	RUS	ROS	SMOTE	ADASYN	SVMSMOTE	SMOTETomek	EasyEnsemble	DiagSelect
Meter C	DT	bAcc	0.809±0.058	0.876±0.039	0.906±0.045	0.902±0.044	0.888±0.035	0.914±0.044	0.915±0.066	0.938±0.065
		Macro- F_1	0.811±0.057	0.883±0.038	0.909±0.043	0.893±0.051	0.885±0.046	0.917±0.04	0.908±0.08	0.94±0.059
		GM	0.797±0.061	0.867±0.043	0.9±0.051	0.899±0.045	0.883±0.039	0.908±0.049	0.911±0.069	0.935±0.067
	SVM	bAcc	0.761±0.082	0.906±0.009	0.901±0.005	0.901±0.005	0.901±0.005	0.913±0.045	0.766±0.077	0.946±0.04
		Macro- F_1	0.762±0.07	0.908±0.009	0.903±0.016	0.903±0.016	0.903±0.016	0.913±0.04	0.764±0.074	0.947±0.039
		GM	0.742±0.082	0.9±0.01	0.894±0.006	0.894±0.006	0.894±0.006	0.909±0.046	0.739±0.087	0.944±0.041
	MLP	bAcc	0.835±0.124	0.955±0.033	0.945±0.049	0.95±0.041	0.93±0.046	0.873±0.025	0.941±0.042	0.976±0.017
		Macro- F_1	0.841±0.122	0.961±0.029	0.949±0.046	0.955±0.037	0.939±0.039	0.861±0.041	0.946±0.037	0.976±0.015
		GM	0.817±0.145	0.952±0.035	0.94±0.054	0.946±0.044	0.927±0.047	0.861±0.031	0.938±0.044	0.974±0.019

TABLE III
DIAGNOSIS PERFORMANCE BY DIFFERENT METHODS IN METER D DIAGNOSTICS DATASET

Dataset	Model	Metric	RUS	ROS	SMOTE	ADASYN	SVMSMOTE	SMOTETomek	EasyEnsemble	DiagSelect
Meter D	DT	bAcc	0.806±0.051	0.889±0.035	0.903±0.049	0.883±0.036	0.879±0.039	0.906±0.044	0.9±0.034	0.916±0.034
		Macro- F_1	0.811±0.058	0.892±0.036	0.905±0.045	0.885±0.036	0.884±0.032	0.909±0.037	0.902±0.03	0.92±0.033
		GM	0.78±0.079	0.883±0.036	0.898±0.051	0.875±0.039	0.872±0.041	0.901±0.045	0.895±0.035	0.912±0.037
	SVM	bAcc	0.89±0.034	0.893±0.046	0.91±0.046	0.906±0.032	0.893±0.046	0.896±0.037	0.914±0.041	0.934±0.019
		Macro- F_1	0.894±0.024	0.899±0.039	0.916±0.036	0.91±0.03	0.899±0.039	0.902±0.037	0.92±0.033	0.938±0.013
		GM	0.878±0.037	0.881±0.055	0.903±0.053	0.896±0.04	0.881±0.055	0.89±0.04	0.907±0.047	0.931±0.021
	MLP	bAcc	0.808±0.048	0.862±0.02	0.866±0.032	0.861±0.066	0.878±0.03	0.849±0.06	0.9±0.028	0.907±0.033
		Macro- F_1	0.812±0.039	0.87±0.024	0.863±0.046	0.865±0.058	0.887±0.021	0.849±0.071	0.896±0.026	0.912±0.032
		GM	0.789±0.05	0.85±0.026	0.856±0.041	0.85±0.073	0.868±0.031	0.834±0.072	0.894±0.031	0.902±0.035

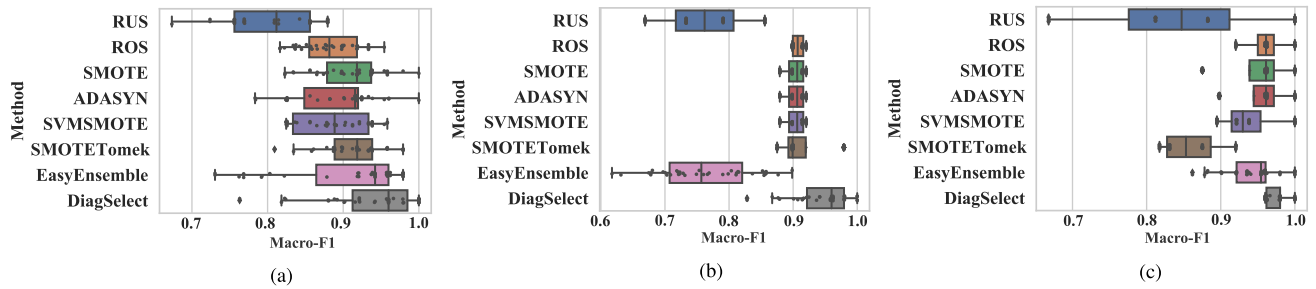


Fig. 5. Boxplot of Macro- F_1 in Meter C diagnostics dataset: (a) DT, (b) SVM, and (c) MLP.

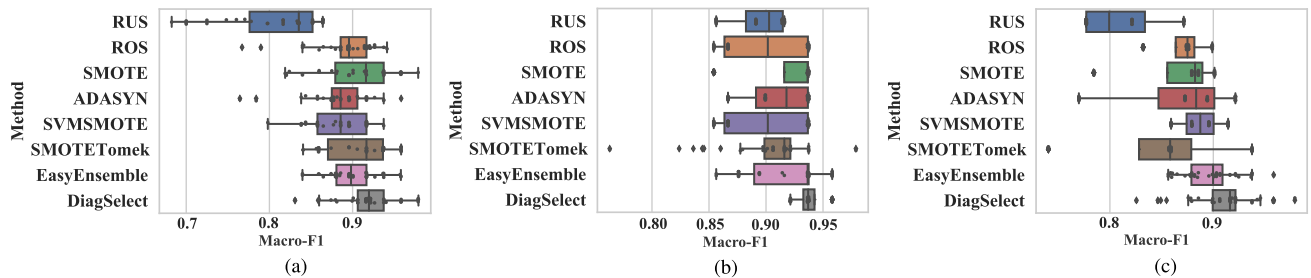


Fig. 6. Boxplot of Macro- F_1 in Meter D diagnostics dataset: (a) DT, (b) SVM, and (c) MLP.

task pairs. The experiment result of transferability is shown in Fig. 7. In the cross-model experiment, the largest relative performance loss (-25.08%) is reported in the train-Meter C (SVM) and test-Meter D (DT) task pair set, which shows that the relative performance loss of cross-dataset and cross-model tasks is the largest in all train-test task pairs. The smallest relative

performance loss (-1.21%) is reported in the Train-Meter D (MLP) and Test-Meter D (SVM) task pair settings, which indicates that the same-dataset and similar cross-model tasks have the smallest relative performance loss. This comparative experiment shows that DiagSelect has transferability in similar datasets and similar models. Even for tasks with different

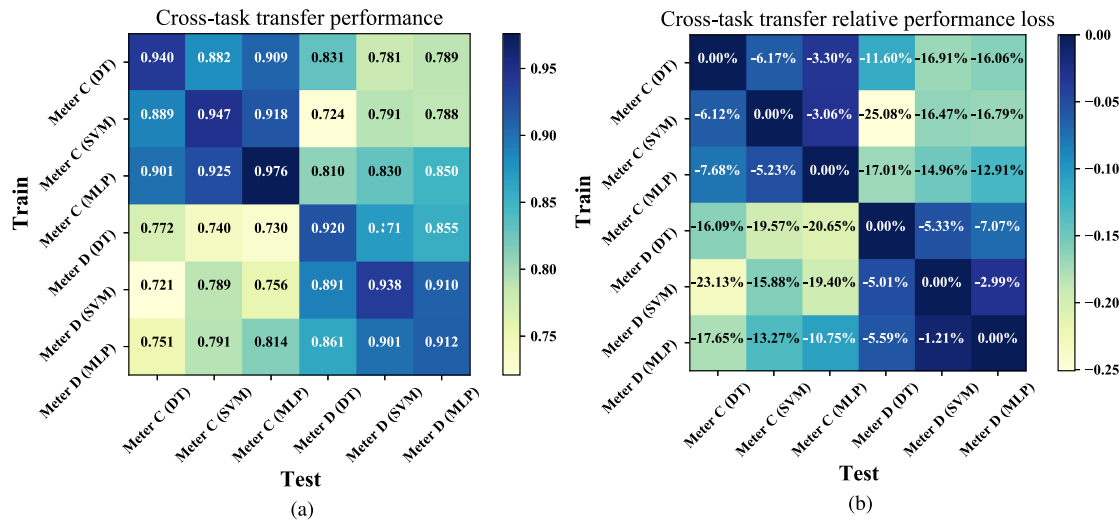


Fig. 7. Cross-task transfer performance of DiagSelect in terms of Macro- F_1 . (a) Cross-task transfer performance. (b) Cross-task transfer performance loss.

datasets and different models, DiagSelect still has transferability. This transferability allows DiagSelect to be used directly or initially for another similar task to reduce the calculation costs.

V. CONCLUSION

In this article, a novel adaptive sample selection method called DiagSelect was proposed to settle the class-imbalance problem in the fault diagnosis domain. Different from the conventional fault diagnosis methods for the class-imbalance problem, DiagSelect was a novel strategy that can select samples adaptively by the reward to improve the diagnosis performance of the diagnosis model. The optimization problem of DiagSelect was formulated as a MAB problem and optimized by REINFORCE. This adaptive automatic process of the sample selection makes the diagnosis model perform better based on the data distribution without excessively increasing expert knowledge and domain knowledge. The proposed DiagSelect was evaluated in the synthetic dataset and the real-world datasets. The experimental results showed that DiagSelect has the following remarkable advantages: 1) DiagSelect has effectiveness and stability for settling the class-imbalance problem and outperforms other state-of-the-art strategies in the fault diagnosis domain; 2) DiagSelect has the characteristic of adaptive sample selection; and 3) the transferability experiment showed that DiagSelect has transferability, so it can be used as an initial model for similar tasks to reduce the calculation costs. As a data level technique, DiagSelect can be easily used in conjunction with the conventional diagnosis models. Thus, DiagSelect can be easily used with arbitrary evaluation criteria and the diagnosis models in the fault diagnosis domain of the industrial process for the class-imbalance problem.

REFERENCES

- [1] M. Kano and Y. Nakagawa, "Data-based process monitoring, process control, and quality improvement: Recent developments and applications in steel industry," *Comput. Chem. Eng.*, vol. 32, no. 1/2, pp. 12–24, 2008.
- [2] Z. Ge, S. Zhong, and Y. Zhang, "Semisupervised kernel learning for FDA model and its application for fault classification in industrial processes," *IEEE Trans. Ind. Informat.*, vol. 12, no. 4, pp. 1403–1411, Aug. 2016.
- [3] O. Briones, R. Alarcón, A. J. Rojas, and D. Sbarbaro, "Tuning generalized predictive PI controllers for process control applications," *ISA Trans.*, to be published, doi: [10.1016/j.isatra.2021.02.040](https://doi.org/10.1016/j.isatra.2021.02.040).
- [4] X. Zhang, C. Wei, and Z. Song, "Fast locally weighted PLS modeling for large-scale industrial processes," *Ind. Eng. Chem. Res.*, vol. 59, no. 47, pp. 20779–20786, 2020.
- [5] Z. Ge, Z. Song, S. X. Ding, and B. Huang, "Data mining and analytics in the process industry: The role of machine learning," *IEEE Access*, vol. 5, pp. 20590–20616, Sep. 2017.
- [6] M. E. Koujok, A. Ragab, H. Ghezzaz, and M. Amazouz, "A multi-agent-based methodology for known and novel faults diagnosis in industrial processes," *IEEE Trans. Ind. Informat.*, vol. 17, no. 5, pp. 3358–3366, May 2021.
- [7] Z. Hu and P. Jiang, "An imbalance modified deep neural network with dynamical incremental learning for chemical fault diagnosis," *IEEE Trans. Ind. Electron.*, vol. 66, no. 1, pp. 540–550, Jan. 2019.
- [8] T. Fei, Q. Qi, A. Liu, and A. Kusiak, "Data-driven smart manufacturing," *J. Manuf. Syst.*, vol. 48, pp. 157–169, 2018.
- [9] I. Mani and I. Zhang, "kNN approach to unbalanced data distributions: A case study involving information extraction," in *Proc. Workshop Learn. Imbalanced Datasets*, 2003, pp. 1–7.
- [10] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "Smote: Synthetic minority over-sampling technique," *J. Artif. Intell. Res.*, vol. 16, no. 1, pp. 321–357, 2011.
- [11] H. He, Y. Bai, E. A. Garcia, and S. Li, "Adasyn: Adaptive synthetic sampling approach for imbalanced learning," in *Proc. IEEE Int. Joint Conf. Neural Netw.*, 2008, pp. 1322–1328.
- [12] G. E. Batista, A. L. Bazzan, and M. C. Monard, "Balancing training data for automated annotation of keywords: A case study," in *WOB*, 2003, pp. 10–18.
- [13] J. M. Johnson and T. M. Khoshgoftaar, "Survey on deep learning with class imbalance," *J. Big Data*, vol. 6, no. 1, 2019, Art. no. 27.
- [14] C. X. Ling, Q. Yang, J. Wang, and S. Zhang, "Decision trees with minimal costs," in *Proc. 21st Int. Conf. Mach. Learn.*, 2004, Paper 69.
- [15] X.-Y. Liu and Z.-H. Zhou, "The influence of class imbalance on cost-sensitive learning: An empirical study," in *Proc. IEEE 6th Int. Conf. Data Mining*, 2006, pp. 970–974.
- [16] P. Peng, W. Zhang, Y. Zhang, Y. Xu, H. Wang, and H. Zhang, "Cost sensitive active learning using bidirectional gated recurrent neural networks for imbalanced fault diagnosis," *Neurocomputing*, vol. 407, pp. 232–245, 2020.
- [17] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 2980–2988.

- [18] B. Krawczyk, "Learning from imbalanced data: Open challenges and future directions," *Prog. Artif. Intell.*, vol. 5, no. 4, pp. 221–232, 2016.
- [19] R. J. Williams, "Simple statistical gradient-following algorithms for connectionist reinforcement learning," *Mach. Learn.*, vol. 8, no. 3/4, pp. 229–256, 1992.
- [20] R. Razavi-Far, M. Farajzadeh-Zanjani, and M. Saif, "An integrated class-imbalanced learning scheme for diagnosing bearing defects in induction motors," *IEEE Trans. Ind. Informat.*, vol. 13, no. 6, pp. 2758–2769, Dec. 2017.
- [21] T. Zhang *et al.*, "Intelligent fault diagnosis of machines with small & imbalanced data: A state-of-the-art review and possible extensions," *ISA Trans.*, to be published, doi: [10.1016/j.isatra.2021.02.042](https://doi.org/10.1016/j.isatra.2021.02.042).
- [22] H. Yi, Q. Jiang, X. Yan, and B. Wang, "Imbalanced classification based on minority clustering smote with wind turbine fault detection application," *IEEE Trans. Ind. Informat.*, vol. 17, no. 9, pp. 5867–5875, Sep. 2021.
- [23] X. Yuan, B. Huang, Z. Ge, and Z. Song, "Double locally weighted principal component regression for soft sensor with sample selection under supervised latent structure," *Chemometrics Intell. Lab. Syst.*, vol. 153, pp. 116–125, 2016.
- [24] C. Li, S. Li, A. Zhang, Q. He, Z. Liao, and J. Hu, "Meta-learning for few-shot bearing fault diagnosis under complex working conditions," *Neurocomputing*, vol. 439, pp. 197–211, 2021.
- [25] Y. Zhuo and Z. Ge, "Auxiliary information guided industrial data augmentation for any-shot fault learning and diagnosis," *IEEE Trans. Ind. Informat.*, vol. 17, no. 11, pp. 7535–7545, Nov. 2021.
- [26] D. G. Zill, *Advanced Engineering Mathematics*. Burlington, MA, USA: Jones & Bartlett Publishers, 2020.
- [27] T. Bühler and M. Hein, "Spectral clustering based on the graph p-Laplacian," in *Proc. 26th Annu. Int. Conf. Mach. Learn.*, 2009, pp. 81–88.
- [28] H. M. Nguyen, E. W. Cooper, and K. Kamei, "Borderline over-sampling for imbalanced data classification," *Int. J. Knowl. Eng. Soft Data Paradigms*, vol. 3, no. 1, pp. 4–21, 2011.
- [29] X.-Y. Liu, J. Wu, and Z.-H. Zhou, "Exploratory undersampling for class-imbalance learning," *IEEE Trans. Syst., Man, Cybern., Part B.*, vol. 39, no. 2, pp. 539–550, Apr. 2009.
- [30] C. Cortes and V. Vapnik, "Support-vector networks," *Mach. Learn.*, vol. 20, no. 3, pp. 273–297, 1995.
- [31] D. Dua and C. Graff, "UCI machine learning repository," School Inf. Comput. Sci., Univ. California-Irvine, Irvine, CA, USA, 2017. [Online]. Available: <http://archive.ics.uci.edu/ml>
- [32] J. R. Quinlan, "Induction of decision trees," *Mach. Learn.*, vol. 1, no. 1, pp. 81–106, 1986.
- [33] S. Haykin, *Neural Networks and Learning Machines*. London, U.K.: Pearson, 2010.



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